

DE Lab8  
Hayden Fuller

Q1:

a)

$$B = a / (1 - a) = .98 / .02 = 49$$

i)  $I_C = 2.107\text{mA}$

$$I_B = (V_{CC} - V_B) / R_B = (5 - .7) / 100\text{k} = 0.043\text{mA}$$

$$I_C = B * I_B = 49 * 0.043\text{mA} = 2.107\text{mA}$$

ii)  $V_{BE} = 0.737783\text{V}$

$$I_C = I_{ES} * e^{(V_{BE} / V_T)}$$

$$V_{BE} = V_T * \ln(I_C / I_{ES})$$

$$V_{BE} = 0.026 * \ln(2.107\text{mA} / 10^{-15}) = 0.026 * 28.376 = 0.737783\text{V}$$

iii)  $V_{BE}$  ON ;  $V_{BC}$  OFF

$$V_{BE} = 0.7377\text{V} > 0.7\text{V ON}$$

$$I_C = (V_{CC} - V_C) / R_C$$

$$V_C = V_{CC} - I_C * R_C$$

$$V_{BC} = V_B - V_C = 0.7377 - (5 - 2.107\text{mA} * 1\text{k}) = 0.7377 - 5 + 2.107 = -2.1553 < 0.5\text{V OFF}$$

iv)  $V_{CE} = 2.893\text{V}$

$$V_{CE} = V_C - V_E = V_{BE} - V_{BC} = 0.7377 - -2.1553 = 2.893\text{V}$$

v)

LTspice

$$I_C = 2.090\text{mA}$$

$$V_{CE} = 2.910\text{V}$$

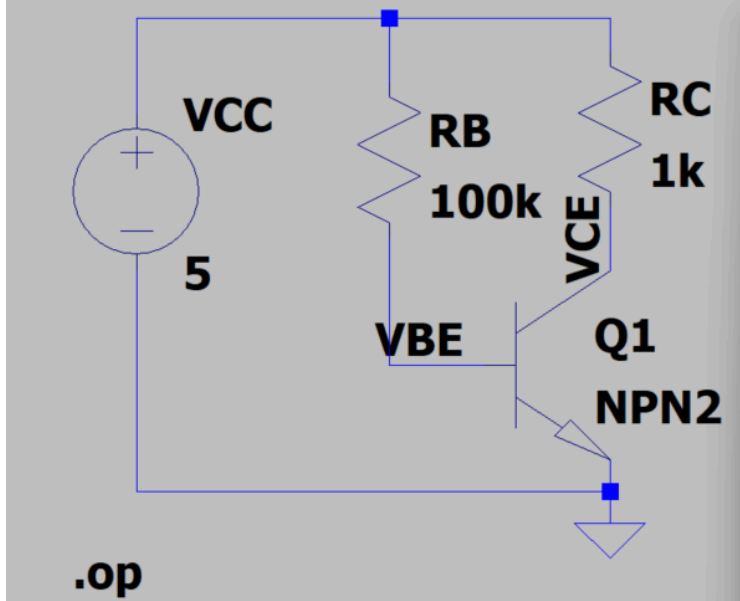
Hand

$$2.107\text{mA}$$

$$V_{CE} = 2.893\text{V}$$

these are very close values, only 17mV or 0.5% off

**.MODEL NPN2 npn(bf=49 br=2 Ibc=1e-15 Ibe=1e-15)**



C:\Users\Hayden\Documents\LTspice\Draft17.net

--- Operating Point ---

|          |             |                |
|----------|-------------|----------------|
| V(n001): | 5           | voltage        |
| V(vbe):  | 0.733748    | voltage        |
| V(vce):  | 2.90954     | voltage        |
| I(RB):   | 4.26625e-05 | device_current |
| I(RC):   | 0.00209046  | device_current |
| I(VCC):  | -0.00213313 | device_current |
| Ib(Q1):  | 4.26625e-05 | device_current |
| Ic(Q1):  | 0.00209046  | device_current |
| Ie(Q1):  | -0.00213313 | device_current |
| Is(Q1):  | 0           | device_current |

b)

$$B = \alpha / (1 - \alpha) = .98 / .02 = 49$$

i)  $I_C = 4.8\text{mA}$

$$V_{BC} = 0.5\text{V}, V_{BE} = 0.7\text{V}, V_{CE} = 0.2$$

$$I_C = (V_{CC} - V_C) / R_C = (5 - .2) / 1\text{k} = 4.8\text{mA}$$

ii)  $V_{BE} = 0.759191\text{V}$

$$I_C = I_{ES} \cdot e^{(V_{BE} / V_T)}$$

$$V_{BE} = V_T \cdot \ln(I_C / I_{ES})$$

$$V_{BE} = 0.026 \cdot \ln(4.8\text{mA} / 10^{-15}) = 0.026 \cdot 29.200 = 0.759191\text{V}$$

iii)  $I_C = 4.8\text{mA} < 21.07\text{mA} = B \cdot I_B$

$$I_B = (V_{CC} - V_B) / R_B = (5 - .7) / 10\text{k} = 0.43\text{mA}$$

$$B \cdot I_B = 49 \cdot .43\text{mA} = 21.07\text{mA}$$

iv) 0.2

from part 1.b.i

v)

LTspice

$$I_C = 4.941\text{mA}$$

$$V_{CE} = 58.540\text{mV}$$

Hand

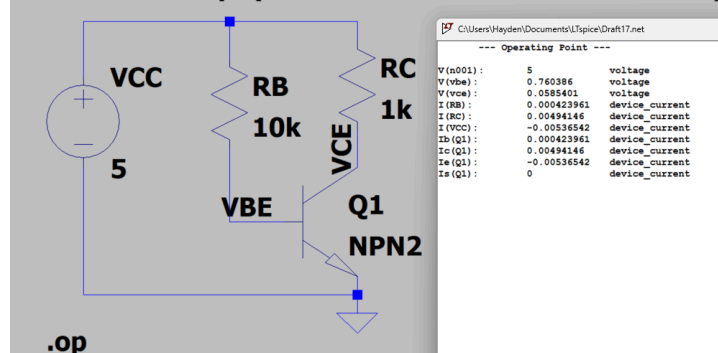
$$4.8\text{mA}$$

$$V_{CE} = 0.2\text{V}$$

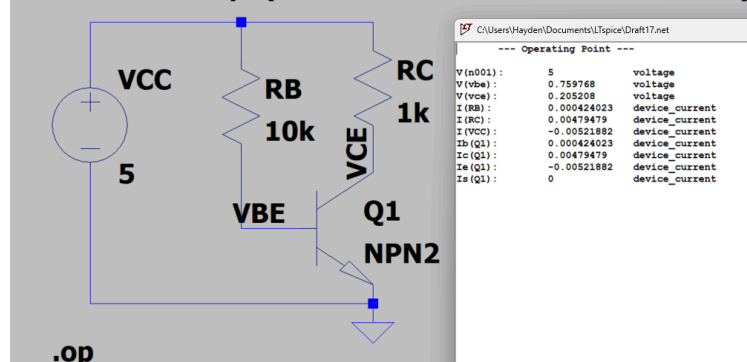
these  $I_C$  values are close, only 0.141mA or 3% off

the  $V_{CE}$  values are off by a factor of 4.

**.MODEL NPN2 npn(bf=49 br=2 Ibc=1e-15 Ibe=1e-15)**



**.MODEL NPN2 npn(bf=49 br=2 Ibc=3e-13 Ibe=1e-15)**



c)

$$I_{ES} = A_E q D_n n_i^2 / (N_E W_B)$$

$A_E$  increase x10,  $I_{ES}$  increase x10

$$I_{ES} = 10^{-14} \text{ A}$$

i)

$$V_{BE} = V_T \ln |I_C / I_{SE}|$$

$$V_{BE} = 0.026 \ln(2.107 \text{ mA} / 10^{-15}) = 0.026 \cdot 28.376 = 0.737783 \text{ V}$$

$I_{SE}$  increase x10

$$V_{BE} = 0.026 \ln(2.107 \text{ mA} / 10^{-14}) = 0.026 \cdot 28.376 = 0.677916 \text{ V}$$

$V_{BE}$  decrease by 0.059867 V